CURRICULUM VITAE



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Jiyang District, Sanya, Hainan, China

PERSONAL

Born 24 April 1991
Shenyang, Liaoning, China

EXPERTISE

Coupled aero- and hydrodynamics of sail propulsion
Simulation-driven multidisciplinary optimization
Autonomous mission planning
Data-driven control



OCEAN ROBOTICS

DESIGN / PLANNING / CONTROL

Yang An

Engineer / Doctor of Engineering

Marine Robotics and Intelligent Navigation

Profile

Engineer at the Institute of Deep-sea Science and Engineering, Chinese Academy of Sciences, with a doctorate in engineering. My research on autonomous sailboats for ocean observation spans system design and optimization, performance evaluation, mission planning, and the transfer of racing sailors' expertise to autonomous navigation.

Research Areas

Technology landscape and cross-domain transfer
Review autonomous sailboat design and technology through performance requirements. Compare modeling, control, and weather routing in racing and robotic sailing to identify synergies and conditions for transferring methods.

Autonomous sailboat design and optimization
Study how rigs, hulls, and appendages work together through velocity prediction, performance evaluation, and simulation-driven optimization. Apply Bayesian optimization and knowledge transfer to overall configuration design and investigate hydrofoil-based stabilization.

Path planning and mission decision-making
Develop global path planners for multipoint AUV seabed surveys. Study mission planning and navigational feasibility for autonomous sailboat coverage in spatially varying, time-dependent wind and current fields.

Data-driven control and policy transfer
Model how racing sailors maneuver and make decisions from data, learn navigation policies, and study how to transfer those policies to autonomous sailboat control.

Education and Appointments

Feb 2026 - Present
Engineer, Department of Deep-sea Information Technology
Institute of Deep-sea Science and Engineering, Chinese Academy of Sciences, Sanya

Sep 2023 - Sep 2025
Postdoctoral Researcher, Control Science and Engineering
Tsinghua Shenzhen International Graduate School, Tsinghua University, Shenzhen
Postdoctoral supervisor: Assistant Professor Zhengru Ren

Sep 2015 - Jun 2023
Combined Master's and Doctoral Program, Mechatronic Engineering
University of Chinese Academy of Sciences and Shenyang Institute of Automation, Chinese Academy of Sciences, Shenyang
Dissertation: *Research on Design and Optimization Methods for the Speed Performance of Autonomous Sailboats*
Doctoral advisor: Prof. Jiancheng Yu
Master's advisor: Prof. Jian Liu

Sep 2010 - Jun 2014
B.Eng., Naval Architecture and Ocean Engineering
School of Navigation and Naval Architecture, Dalian Ocean University, Dalian

Publications

02

Journal Articles

01 **Yang An**, Jean-Baptiste R.G. Souppez, Zhikang Ge, Bo Peng, Mengwei Zhang, Gaofei Xu, Zhengru Ren*
Sailing forward: A review of contrasts and synergies between racing and robotic sailing
Journal of Ocean Engineering and Science · 2026 · 11(5) · 1700-1722
DOI 10.1016/j.joes.2026.03.008

02 Taiyu Zhang, Can Ma, **Yang An**, Zhengru Ren*
Estimating semi-stationary sea states through wavelet-based adaptive segmentation of ship motions
Ocean Engineering · 2026 · 350 · 124030
DOI 10.1016/j.OCEANENG.2025.124030

03 **Yang An***, Feng Hu, Kuo Chen, Jiancheng Yu*
Optimising General Configuration of Wing-Sailed Autonomous Sailing Monohulls Using Bayesian Optimisation and Knowledge Transfer
Journal of Marine Science and Engineering · 2023 · 11(4) · 703
DOI 10.3390/jmse11040703

04 **Yang An**, Jiancheng Yu*, Feng Hu, Zhenyu Wang
Towards a general design evaluation tool: The development and validation of a VPP for autonomous sailing monohulls
Applied Ocean Research · 2022 · 120 · 103053
DOI 10.1016/j.apor.2022.103053

05 **Yang An**, Jiancheng Yu*, Jin Zhang
Autonomous sailboat design: A review from the performance perspective
Ocean Engineering · 2021 · 238 · 109753
DOI 10.1016/j.oceaneng.2021.109753

Conference Papers

06 Bo Peng, **Yang An**, Mengwei Zhang, Zhengru Ren
High-Fidelity Process Plant Modeling of an Unmanned Sailboat with a Wing Sail Based on Multi-body Dynamics
IFAC-PapersOnLine · 2025 · 59(22) · 525-530
DOI 10.1016/j.ifacol.2025.11.687

07 Mengwei Zhang, Zhengru Ren, Yuqing He, Decai Li, Liying Yang, **Yang An**
A Coordinate Coverage Water Area Monitoring Scheme for Multiple Unmanned Surface Vehicles in Dynamic Environments
2025 IEEE International Conference on Unmanned Systems (ICUS) · 2025 · 1695-1700
DOI 10.1109/ICUS66297.2025.11295601

08 Can Ma, Taiyu Zhang, Zongyuan Yang, **Yang An**, Xiang Yuan Zheng, Zhengru Ren
Relative Motion Prediction for Integrated Floating Offshore Wind Turbine Installation Scheme Based on Data-Driven
Volume 7: Ocean Renewable Energy - ASME 2024 43rd International Conference on Ocean, Offshore and Arctic Engineering · 2024
DOI 10.1115/OMAE2024-124558

09 **Yang An**, Gaofei Xu, Chunhui Xu, Hongyu Zhao, Jian Liu
Path Planning for Multipoint Seabed Survey Mission Using Autonomous Underwater Vehicle
OCEANS 2017 - Anchorage · 2017 · 1-5
https://ieeexplore.ieee.org/document/8232284

Research and Engineering



Grants and Research Projects

Dec 2023 - Dec 2025

Full-scale Data-driven Modeling and Control of Soft-sail Sailboats / Principal Investigator
Shenzhen Municipal Higher Education Stable Support Program (General Project) | RMB 100,000
Designed and commissioned a data acquisition platform for racing sailboats and established an end-to-end collection workflow. Partnered with sailing schools for data access and field sites. Developed a dynamic model of an autonomous sailboat.

Apr 2025 - Aug 2025

Key Technologies for a Digital Smart Dry Dock Based on Floating Offshore Structures / Research Team Member
Newly established technology project, China National Offshore Oil Corporation (CNOOC)
Conducted feasibility studies and background research, prepared the proposal and presentation, and coordinated requirements with partner organizations.

Sep 2023 - Nov 2023

Key Technologies for a 4,000-m-Class Multipurpose Tethered ROV for Deep-Sea Resource Development / Research Team Member
Shenzhen Major Science and Technology Program
Evaluated the overall system concept and prepared the project presentation.

Oct 2018 - Oct 2020

Novel Hydrofoil-Based Autonomous Sailboat / Principal Investigator
Special research program | RMB 1,000,000
Led overall sailboat design, developed a method for evaluating system performance, and designed and optimized the hull and hydrofoil geometries.

Mar 2019 - Feb 2020

Feasibility Study for a Research Facility for Multiscale Autonomous Observation of Extreme Ocean Dynamics / Research Team Member
National Development and Reform Commission pre-research program for major scientific facilities
Evaluated hull concepts, the platform general arrangement, and survivability under extreme marine conditions.

Mar 2017 - Mar 2018

Explorer 4500 AUV System for Deep-Sea Hydrothermal Exploration / Research Team Member
Strategic Priority Research Program, Chinese Academy of Sciences
Developed and tested algorithms for the 2D multibeam forward-looking sonar module in a hardware-in-the-loop simulator. Designed a mission-level global planner for AUV seabed surveys.

Engineering Competitions

Oct 2020 - Dec 2020

Zhuhai Wanshan International Intelligent Vessel Competition / Team Leader
Southern Marine Science and Engineering Guangdong Laboratory (Zhuhai)
Designed and implemented camera- and LiDAR-based perception. Led path planning and competition strategy. Won USD 20,000 in prize money.

Feb 2018 - Jun 2018

1st Pioneer Cup Future Technology Innovation Competition / Team Leader
Chinese Academy of Sciences
Researched, prepared, and pitched the novel hydrofoil-based autonomous sailboat project. The project was approved, and the team won RMB 30,000.

Oct 2016 - Dec 2016

Liaoning Provincial Graduate Big Data Innovation Competition / Team Leader
Liaoning Provincial Department of Education
Designed and simulated a school bus routing algorithm for primary and secondary students, prepared the presentation, and pitched the solution.

Mar 2016 - Jul 2016

RoboMasters 2016 National Undergraduate Robotics Competition / Vision Team Lead
School Department of the Communist Youth League Central Committee
Led visual recognition, tracking, and control for base defense and the large power-rune target. Placed third in the Eastern Division, won RMB 10,000, and earned USTC a seeded place in the next competition.

Current Work and Service

04

Preprint and Manuscripts

01 **Yang An**, Zhikang Ge, Taiyu Zhang, Jean-Baptiste R. G. Souppez, Gaofei Xu, Zhengru Ren

Coverage path planning for autonomous sailboats in inhomogeneous and time-varying oceans: A spatiotemporal optimization approach

arXiv preprint · 2026

https://arxiv.org/abs/2602.15901

02 **Yang An**, Zhengru Ren, Shenna Wu, Kuo Chen, Shanyu Zhong

Development and application of a data collection system for soft-sail sailboat maneuvering behavior

In preparation

03 Zhier Chen, Kuo Chen, Minghao Dou, Qianqi Liu, **Yang An**, Kai Ren, Zeming Wu, Yu Tian, Jie Sun, Xinping Wang, Jiancheng Yu

Dynamic modeling, parameter identification and numerical analysis of cables in flexibly connected dual-AUV systems

Submitted to Mechanical Systems and Signal Processing

Honors and Awards

2022	2nd Pioneer Cup Future Technology Innovation Competition	Team Advisor	Merit Prize (semifinal)
2020	Zhuhai Wanshan International Intelligent Vessel Competition	Team Leader	Overall Champion and Obstacle Navigation Event Champion
2018	1st Pioneer Cup Future Technology Innovation Competition	Team Leader	Merit Prize (semifinal)
2016	Liaoning Provincial Graduate Big Data Innovation Competition	Team Leader	First Prize
2016	RoboMasters National Undergraduate Robotics Competition	Vision Team Lead	Eastern Division third place and national top 32

Graduate Mentoring

2025 master's cohort — **Sicheng Yang** / University of Chinese Academy of Sciences
Mechatronic Engineering, joint supervision
Acoustic-image-based underwater pipeline recognition and AUV tracking

2025 master's cohort — **Yongzhen Jia** / Shenyang University of Technology
Artificial Intelligence, joint supervision
Physics-informed simulation and reconstruction of flexible sail shapes

Professional Service

Editorial Board Member — Journal of Sailing Technology

Journal Reviewer — Ocean Engineering, Structural and Multidisciplinary Optimization, and Journal of Sailing Technology

References

Zhengru Ren Associate Professor	School of Naval Architecture, Ocean and Civil Engineering, Shanghai Jiao Tong University	ren.zhengru@sjtu.edu.cn
Jiancheng Yu Professor	College of Ocean and Earth Sciences, Xiamen University	jcyu@xmu.edu.cn
Jin Zhang Professor	School of Physics, Liaoning University	jinzhang@lnu.edu.cn
Wei Guo Research Professor	Institute of Deep-sea Science and Engineering, Chinese Academy of Sciences	guow@idsse.ac.cn